

PDF file name: Elsevier-2021-Towards security automation in Software Defined

Original article word count: 12618

Extracted content word count: 2589 (20% of the original word count)

Title of the article: *Computer Communications*

The content extracted is as follows:

1. Introduction

Security demands in new complex and dynamic networks, such as IoT [1–4], cloud computing [5], 5G/6G communication networks [6,7], and vehicular communications, [8,9], in addition to the increasing number of sophisticated and intelligent attacks, are key motivating factors for deploying automated defense solutions for networks using the Software-Defined Networking (SDN) technology.

Including automation in the heart of network security management will help to deal with the increasing number of alerts and skilled IT administrators' shortages.

SDN is enabling security managers to bridge the response resource gap through intelligent incident detection and automated response.

This networking technology presents new security challenges for network designers and operators, but it also provides a platform for implementing reliable, centrally managed, and automated security solutions for networks.

Although many surveys have already studied intelligent methods applied in SDN-based security solutions, none have comprehensively analyzed and compared the complexity and level of automation achieved with these techniques.

The idea of security automation in SDN environments is fundamental since this technology was created to facilitate network operation and management with minimal human intervention, which is considered error-prone.

In this paper, we survey existing research efforts related to the automation of security in SDN-based networks, using peer-reviewed studies published in the last seven years (2015–2021).

The methods of AI have been used to discover threats in networks [23–25] since they are considered as robust potential solutions to make networks self-aware, self-adaptive, self-secured, self-healing, and self-managed [26,27].

The study of SDN-based reactive security solutions requires an analysis of the characteristics that makes detection and mitigation mechanisms adaptive to changes in the network.

On the other hand, proactive security solutions use emerging concepts such as Network Function Virtualization, Moving Target Defense, Cyber Deception, Network Slicing, and Blockchain to secure the SDN architecture effectively.

The contributions of this work are: This work represents the first attempt to study SDN-based security solutions from an automation perspective.

These reviews divided security and SDN into two categories: security for SDN, and SDN as a security solution for networks.

In the first category, security issues specific to the SDN architecture and their solutions have been surveyed [28–31].

The most common security issues targeting the SDN elements are the DoS/DDoS attacks.

In the second category, the SDN-based security solutions to protect traditional networks have been reviewed. The standard strategy to review these solutions has been to divide them into statistics-based, ML-based, DL-based, and metaheuristics-based solutions [23–25,36–38].

These reviews define different types of existing threats to networks and study SDN-based solutions to these threats.

Besides, several surveys integrate the study of both categories, security for SDN and SDN-based security solutions [26,42,43].

In contrast to previous surveys, we study SDN-based security solutions for networks from an automation point of view.

No previous survey has studied the automation capabilities of SDN-based security solutions.

We also introduce a discussion about the scalability and assessment factors of the SDN-based security systems, which are fundamental in the design of complete, trustworthy, and ready to deploy automated solutions.

Finally, we present the problems and challenges of current defense solutions and future research directions in the area of security automation in SDN environments.

Section 4 reviews the reactive security solutions for networks using SDN and intelligent techniques.

2. An overview of SDN

To implement novel solutions, such as traffic management in high-performance networks, one has a difficult task since it results unfeasible due to the bottleneck created in the controller and the switches.

These solutions can be seen as alternatives to Open SDN, which address real-life problems.

Table 2 shows these alternatives to SDN-like solutions [45].

It is important to recognize that these alternative SDN-like solutions do not face all the issues and implementation challenges of the Open SDN.

For instance, openness is not a characteristic of the examples of SDN via hypervisor presented in Table 2. In this survey, we include both Open SDN and SDN-like based security solutions.

3. Defense automation in SDN-based networks

Network security automation is a prominent area of study and is expected to be solved by modern network architectures, such as SDN.

Proactive solutions prevent attacks by leveraging techniques such as Moving Target Defense, Network Function Virtualization, Cyber Deception, Network Slicing, and Blockchain technology.

Fig. 4 shows the most used ML methods to implement SDN-security solutions: Support Vector Machine (SVM) [57–59], -Nearest Neighbors (KNN) [60,61], Random Forest (RF) [56], Artificial Neural Network (ANN) [62], Fuzzy Logic [63], and Hidden Markov Model (HMM) [64, 65].

Most of ML-based security solutions focus on Network/Transport Layer (L3/L4) DDoS attacks, which are high-volume attacks and easier to detect.

Most of the ML-based solutions have been tested using security datasets or small test networks.

However, in these solutions, attackers still use part of the network resources.

A need arose for a defense system that would not only detect an intrusion on time but also would make the most optimal real-time crisis-action decision on how the network security policy should be modified to mitigate the threat.

Nevertheless, solving these issues will lead to a big step towards security automation in networks.

Adversarial learning jointly with SDN technology has not been widely explored to solve the security problem in networks.

5. SDN-based proactive defense

SHIELD applied NFV/SDN for virtualization and dynamic placement of virtualized security functions in the network.

Additionally, works in [106,114] surveyed other recent SDN/NFV based security approaches.

NFV-based security solutions provide ondemand network programmability, energy efficiency, and mobility support, characteristics that are relevant to IoT networks.

The nature of VPP-enabled VNFs will provide the open-access, software-based, scalable, and highperformance capabilities to deploy automated security solutions in SDN-based networks.

Therefore, many challenges remain unsolved, such as optimal selection and deployment of SDN/NFV-based security mechanisms and optimal orchestration over different network domains [3].

The capabilities of SDN provide an opportunity to design an optimized and autonomous solution for network defense, which can permit advanced use cases of the deception platform

such as integrated, proactive threat hunting, and active attacker engagement [120].

This security framework can simulate physical processes operations of Industrial Control Systems to capture intricate cyber physical attacks using SDN technology.

This solution protects key nodes based on game theory (GT) through the formulation of a multistage game by using the architectural advantages of SDN.

Work in [126] presents a cyber deception solution that implements defenses against penetration attacks using a decoy chain development (DCD) method based on the combination of NFV and SDN.

In this defense architecture, the SDN controller monitors the security status of the whole network and can detect possible attack sources using Intrusion Detection System (IDS) technology.

Deception mechanisms have demonstrated the potential to improve the security automation in networks but, at the same time, optimizing the resources.

Network slicing (NS) Network slicing strategies promise effective solutions such as dynamic QoS/QoE management [27] for IoT and 5G/6G [7,129] technologies.

Network slicing allows to isolate different traffics to maintain security and reliability in the network [3].

Thus, individual network applications and services can run on each slice, with different levels of security required by individual slices.

The combination of SDN and NFV on network slicing can support the design of advanced security solutions with fined-grained security features, such as perapplication traffic flow [129,130].

Network slicing, combined with SDN and NFV, promises to enable all self-* capabilities needed to obtain a fully automated solution.

Furthermore, all techniques previously studied, including reactive and proactive solutions, can be integrated into each slice, thus providing a comprehensive security solution for new generation networks.

Self-optimized security solutions using blockchain and SDN require further development.

Table 3 shows a comparison of SDN-based proactive security solutions described before.

6. Analysis and comparison

Complexity and level of automation Table 4 ranks proactive and reactive SDN-based security solutions, which summarizes the findings of the complexity and level of automation of the classes analyzed previously.

In the case of complexity and adaptability capabilities, we provide a Null, Low, Medium, High, and Advanced ranking to assess how the SDN-based security solutions stack up against one another.

Also, we note that the reactive solutions have been more explored than the proactive solutions, which still lacks the study of an optimal trade off between performance and accuracy on defending SDN-based networks.

Techniques of RL, adversarial learning, MTD, cyber deception, and network slicing cannot provide a self-optimized or self-configured security solution without NFV, which provides the flexibility to deploy (virtual) network functions on-demand and in real-time.

Although we can implement all the studied defense solutions through all types of SDN-enabled networks, their complexity of implementation Proactive solutions occur more proper to protect new generation networks, such as IoT, 5G, 6G, vehicular ad-hoc networks (VANETS), unmanned aerial vehicle (UAV) networks [148], etc. These networks have in common that they are heterogeneous, dynamic, and large-scale, making them hard to meet the strict requirements, such as low latency, high reliability, high mobility, top security, and enormous connectivity.

The heterogeneity of the devices could hinder the use of reactive solutions in these networks since they can expose to numerous FPs.

Examples of these combinations are MTD-NFV, MTD Cyber Deception, NF-Network Slicing, MTD-NFV-Cyber Deception, and Blockchain assisted-NFV-Network Slicing-NFV, as shown in Table 3. Finally, we highlight that proactive and reactive solutions can be combined for their application in different scenarios.

7. Discussion

We studied reactive and proactive strategies that pursue security automation in SDN-based networks.

Worthwhile approaches can combine reactive and proactive strategies to automate the security solution in SDN-based networks.

The SDN-based security solution for networks is expected to be automated in the coming years.

Such solutions must consider all advantages and limitations of the SDN architecture.

A solution proposal that does not consider the effects on the network performance, such as the bottleneck to collect packets for analysis, will not be scalable.

A good practice is to use a standard evaluation system of security solutions for networks based on the SDN paradigm.

Next, we analyze the scalability and assessment factors, which we consider essential elements to different security proposals before their implementation in production networks.

Scalable SDN-based security solutions Most of the SDN-based security solutions revised throughout this survey propose to perform main processes in the AP.

That is, as the traffic demand of modern networks increases, analyzing all of these packets with a security system at wire speeds will be challenging using sophisticated algorithms, ensuring at the same time high accuracy.

Effective SDN-based security solutions require considering the optimization of different resources to provide scalability.

Fig. 6 shows specific strategies to add scalability to SDN-based security solutions, which we discuss in the following.

Distributed SDN deployment Distributing monitoring capability in a multi-controller environment [160 162] could also enable the scalability of an SDN-based security solution.

Besides, in network environments that require diversity in SDN controllers, such as in IoT, having a backup controller for each controller may become costly, thus resulting in non-scalable and complex solutions [163].

The emerging NFV technology can enable the scalability and flexibility factors of SDN-based security solutions since NFV deploys different virtual functions (VFs) on-demand and on the fly.

Evaluation of SDN-based solutions It is tough to find many studies with an evaluation that is both comprehensive and realistic.

The most common metrics used to validate SDN-based security solutions for networks, particularly those based on the threat detection/mitigation approach, are accuracy, recall, F-score, FPs rate, ROC curve, and detection rate.

Besides using the classification metrics, we can evaluate the SDN-based security solutions using other performance metrics, as described in the following.

Table 6 defines a set of parameters and metrics to evaluate the network security design using SDN effectively.

The flexibility parameter is also significant since integral security solutions will provide a more complete and holistic system to defend the network.

Direct metrics to assess the scalability property of a security solution include the end-to-end communication delay, the controller CPU and memory usage, the switches TCAM usage, the throughput, and the additional traffic on the network.

A trade off between the large-term advantages of a security solution design and its associated costs will define whether to implement this solution in a specific production network.

Open problems SDN-like architectures can certainly automate different solutions in modern network environments.

Finally, this report highlighted that the current best investment areas in networking trends are AI and security.

A functional and automated security solution must recognize that new network environments manage massive traffic and devices and provide new services with different security requirements.

In addition, there is a need of more studies analyzing the practicality of SDN-based security solutions for partial SDN deployment, also known as hybrid networks [165,166], and design methods to be effective in such a situation.

In summary, P4 will boost SDN, NFV, and Blockchain technologies to create more sophisticated solutions required to automate the network defense, maintaining the desired scalability, simplicity, low costs, and openness.

We can shortly expect intelligent data planes that boost open, high-performance, and automated networks security solutions.

Orchestration for a comprehensive security solution The SDN-based security solutions presented in this survey try to solve a single security threat without manual intervention.

Therefore, implementing isolated security solutions to automate network defense could result in redundant functions in many of their subsystems (such as collecting information), the execution of inconsistent tasks, or conflicts with other non-security-related network applications.

The SDN controller can execute the automated security solutions in a purposeful order and verify the success of each task.

Besides, integrating all network security solutions in a unique framework, one can achieve autonomy and resilience in the overall network operation.

In short, an SDN-based security orchestrator can enable an automated, robust, and holistic defense solution with optimized resources utilization.

The SFC and NFV technologies integration in SDN-based architecture can help us to provide a holistic and self-checking security framework [171,172], which has optimal resources utilization, even in multi-domain environments [173].

Intent-based networking (IBN) is another tool enabling SDN-based security orchestration, which can continuously enforce our defense policies while optimizing network resources.

Thus, IBN simplifies the implementation of SDN-based security solutions and their remediation and assurance.

Moreover, IBN can integrate SDN, NFV, and AI to provide fully automated solutions considering the network state, the dynamic state of all users and applications, the service performance and security requirements, and the business needs across all domains.

9. Conclusion

There are multiple approaches to solve the security problem in SDN-based networks.

The nature of programmable and centralized control of SDN enables this technology to generate scalable, optimized, intelligent, and automated security solutions to defend networks.

We categorized the SDN-based security reviewed into two classes: reactive defense based on attack detection and proactive protection based on in-line network reconfiguration.

The state-of-the-art proactive security solutions implement mechanisms like MTD defense, Cyber Deception, Network Slicing, and Blockchains combined with NFV capabilities and intelligent algorithms.

With this evaluation, we can identify the advantages, limitations, and associated costs, of a security solution and decide whether to implement it in specific production networks.

We hope that our discussions and exploration give readers a profound grasp of how the SDN paradigm is pushing the creation of innovative and automated security solutions for next-generation networks, which we believe is the ultimate goal of SDN technology.